Supplementary material: Computational assessment of EBV-self peptide similarities across HLA class II contexts

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1 Supplementary Methods

1.1 SM1. Frozen-core input and TCR-facing BLOSUM62 score

The initial 6,400-comparison V3 matrix stores one declared P1–P9 core for each peptide arm. These cores were frozen before the score matrix was ranked and were not replaced by the later audit-predictor calls. The initial screen is therefore conditional on the declared-core input. The archival V3 manifest must identify the original core-calling method, version, settings, and tie rule so that this input can be independently regenerated.

For two declared nine-residue cores a and b, the TCR-facing normalized BLOSUM62 score was calculated at $I = \{\text{P2}, \text{P3}, \text{P5}, \text{P7}, \text{P8}\}$ as $S(a,b) = \sum_{i \in I} B(a_i, b_i) \,/\, \sum_{i \in I} \max\{B(a_i, a_i), B(b_i, b_i)\}$. B denotes the published BLOSUM62 substitution value. The numerator sums the five matched residue-pair values; the denominator sums the larger of the two residue-specific self-substitution values at each position. Exact identities are handled by the same formula, and cores containing nonstandard or ambiguous amino-acid symbols were excluded during curation. The score is a relative, register-conditional prioritization quantity, not a probability of presentation, recognition, cross-reactivity, or molecular mimicry.

1.2 SM2. Expanded cohort construction and provisional-tier rules

1.2.1 Expanded 49-pair cohort selection

The expanded 49-pair cohort was analyzed separately from the initial eight-pair evidence audit. All final pair identifiers matched entries in the original 6,400-comparison V3 matrix. The frozen source pool comprised eight Stage-1 evidence-audit candidates, ten previously retained near-miss candidates, and 31 rank-based prefilter candidates from the V3 matrix. Deduplication by allele, declared viral core, and declared self core reduced these 49 source rows to 47 unique pair definitions. Two deterministic replacements were then selected from the relevant within-allele V3 rankings to restore the fixed 49-pair cohort; each replacement had pre-existing viral- and self-arm binding percentile ranks $\leq 20\%$. The final cohort contained seven HLA-DRB1*03:01 pairs, 14 HLA-DRB1*08:01 pairs, 16 HLA-DRB1*13:03 pairs, and 12 HLA-DRB1*15:01 pairs, corresponding to 98 peptide arms. Thus, the expansion increased candidate coverage while retaining the same four alleles represented in the initial eight-pair audit.

1.2.2 Secondary provisional-tier rules

Secondary provisional strata were assigned after the expanded 49-pair screen as routing labels for subsequent experimental binding and register-mapping work. A strict peptide arm required all three binding-predictor percentile

ranks to be $\leq 20\%$, agreement of at least two predictors on the same nine-residue core, and agreement between that core and the declared core. A near-passing arm required declared-core agreement and at least two of three predictor percentile ranks $\leq 20\%$. Computational stratum A (formerly "strong-medium") required strict support for both arms plus exact-sequence, exact-allele direct-binding evidence for at least one arm. Computational stratum B (formerly "weak-medium") required either strict support for both arms without verified exact-allele binding or ligand evidence, or support for both declared registers across two calibrated same-allele PANDORA templates and expected/adjacent-register decoys despite one narrow predictor discrepancy. Computational stratum C (formerly "strong-low") required strict support for one arm and near-passing support for the other. Candidates failing these conditions were left untiered. No candidate received high-priority status because the automated public-evidence endpoint was unavailable under the frozen review protocol. These strata do not establish peptide-HLA affinity, natural presentation, T-cell recognition, cross-reactivity, molecular mimicry, or disease relevance.

1.3 SM3. Statistical calibration and sensitivity analyses

Statistical methods were implemented to test whether observed TCR-facing BLOSUM62 scores exceeded scores expected after disruption of residue-position alignments. Each HLA allele had its own separate simulation with a starting universe of 40 EBV cores and 40 self cores, making 1,600 pairwise comparisons. Ten thousand simulation replicates were completed independently for each allele at random seed `20260914`. These peptides' positions were classified as HLA anchor positions (P1, P4, P6, P9) or designated TCR-facing positions (P2, P3, P5, P7, P8). When making the replicates, HLA anchor positions were held in place while TCR-facing residues were independently permuted within each 9-mer core and each peptide's exposed-residue composition was unchanged by this within-core permutation. Furthermore, each unique peptide was shuffled once per replicate and was then reused across 40 pairings to preserve the Cartesian dependence structure. Then, a recalculation was performed with the unchanged normalized BLOSUM62 score function, with higher scores being ranked higher, and five prespecified outputs were collected: pair-specific upper-tail Monte Carlo p, Benjamini-Hochberg q across eight audited pairs, within-allele maxT family-wise p, the null median, and the 95% null interval. Random re-pairing was excluded because the observed 40×40 matrix already contained every possible pairing.

Next, binding-threshold sensitivity of the frozen 49-pair panel was evaluated using the predicted binding percentile ranks generated by the IEDB-recommended binding method, NetMHCIIpan 4.3 EL, and MixMHC2pred 2.1 context. The specific percentile thresholds evaluated were 1%, 2%, 5%, 10%, and 20%, and the binding requirements were all three predictors passing for both peptide arms or at least two of three predictors passing for both arms. Alongside the binding rules, two register criteria were evaluated as separate grid conditions: each predictor's predicted core start was required either to be identical to the declared core start or to have an absolute start-position difference of ≤ 1 residue. The exact computational identity, version, access date, and independence assessment of the IEDB-recommended method must be supplied with the frozen predictor records; it should not be interpreted as an independent biological measurement.

The threshold grid showed that under the locked all-three-predictor, both-arm rule, the two BALF5-TALDO1 pairs passed only at 20%; neither passed at 10% or below. Under the two-of-three, both-arm rule, the DRB1*15:01 pair passed at 10%, whereas the DRB1*13:03 pair first passed at 20%. The ± 1 register rule added no candidates to the strict all-three-predictor condition; it added one nonlead candidate only in the relaxed two-of-three 20% condition. Thus, the two-pair strict result is conditional on the 20% threshold.

Four-metric rank sensitivity was also performed on the four allele-specific cohorts containing 1,600 pairs each. Pairs were evaluated using TCR-facing BLOSUM62, full-core BLOSUM62, TCR-facing sequence identity, and Grantham distance. Higher similarity was better for BLOSUM62 and identity, while lower Grantham distances indicated greater physicochemical similarity. Standard Grantham distance [9] was used as the physicochemical metric. The pairwise score was calculated by averaging the published positional distances between matched TCR-facing residues at P2, P3, P5, P7, and P8 in the two peptide cores. The control performance of standard Grantham distance was tested across eight control panels representing three independent TCR systems, with comparison

based sequentially on the number of systems with a positive-control rank of ≤ 3, the worst system rank, and the system-weighted mean reciprocal rank. Only after Grantham passed this control gate was its use permitted in the four-metric rank sensitivity. For a candidate pair to be robust, it needed to be within the top 1% under at least three metrics, while partially robust pairs did not meet the robust criterion but were within the top 5% under at least three metrics. For tie handling, average ranks were used to calculate pairwise within-allele Spearman correlations when ties occurred, and competition ranks were used for primary robustness labels. The top 1% of pairs was defined as rank ≤ 16, and the top 5% was defined as rank ≤ 80.

1.4 SM8. Calibration and exploratory modeling

The literature-derived Hy.2E11 cross-reactive benchmark, $BALF5_{627\text{-}641}$-$MBP_{85\text{-}99}$, served as a structural-comparison reference [11]. The benchmark contains HLA-DRA*01:01-HLA-DRB5*01:01 for BALF5 (PDB: 1H15) and HLA-DRA*01:01-HLA-DRB1*15:01 for MBP (PDB: 1BX2). It was therefore treated as a DR15 haplotype-level control, rather than a same-allele comparison; it does not validate either BALF5-TALDO1 pair or establish biological mimicry.

Poisson-Boltzmann/APBS pMHC surface-potential comparison was explored as an exploratory surface-resemblance descriptor [12]. Its score was excluded from primary ranking because a 30-result control gate yielded 15 failed and 12 missing results. This did not test or refute biological T-cell cross-reactivity. An AlphaFold 3 TCR-pMHC calibration against 1YMM and 2WBJ was also assessed with DockQ and CAPRI pose-recovery criteria [13, 14]; all ten returned models scored below the DockQ acceptable-quality boundary of 0.23 and both controls were CAPRI "Incorrect," so these outputs were excluded from candidate TCR ranking.

Separately, TCRModel2 generated five-component TCR-pMHC models comprising HLA-alpha, HLA-beta, peptide, TCR-alpha, and TCR-beta chains [15]. The calibration references 1YMM, 1ZGL, and 2WBJ each received five ranked models. DockQ and CAPRI classifications were evaluated alongside TCR-placement, pMHC, TCR-fold, and whole-complex RMSD. 1ZGL produced five acceptable models (DockQ 0.411-0.479); 1YMM produced five incorrect models (0.010-0.140); and 2WBJ produced five incorrect models (0.016-0.038). Only one control met the recovery criterion. Candidate outputs were excluded from ranking and cannot support shared-TCR or cross-reactivity claims; BALF5 replicate models remain exploratory internal predictions rather than docking validation.

2 Supplementary Figures

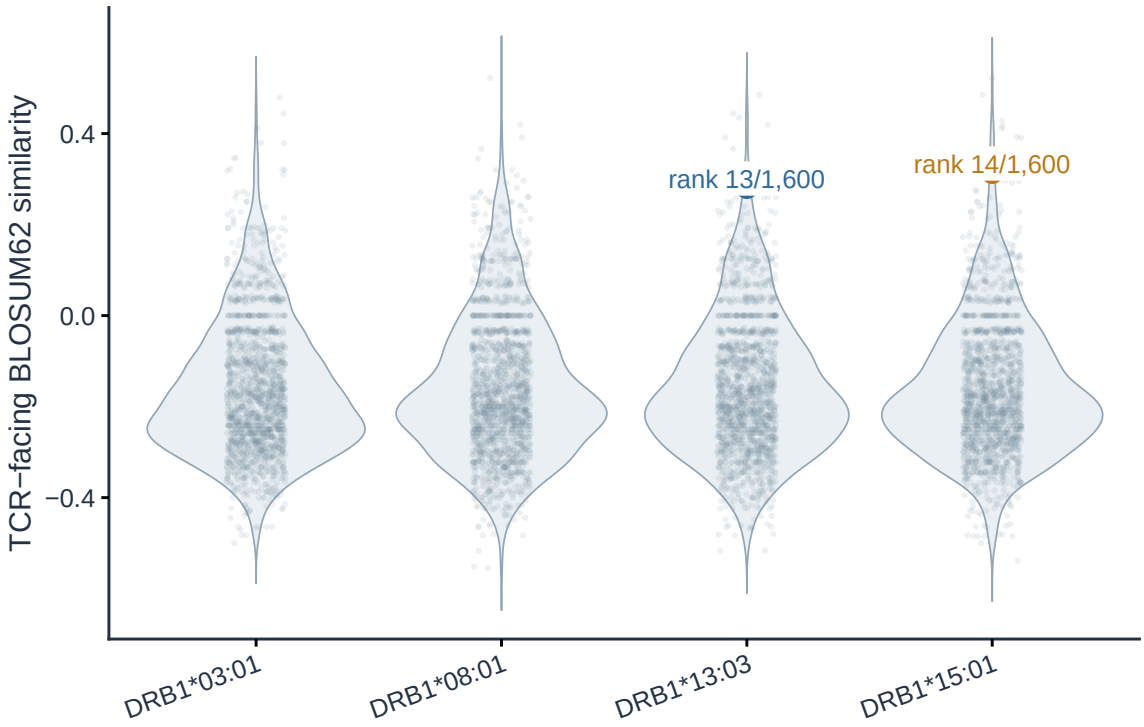


Figure S1: **Within-allele distributions of TCR-facing BLOSUM62 similarity scores.** Each point represents one EBV-human peptide comparison ($n = 1{,}600$ per HLA allele); violin profiles summarize the allele-specific score distributions. Labels identify the BALF5-TALDO1 pairs selected for evidence review. Reported ranks are calculated within each allele, not across the combined 6,400-pair dataset.

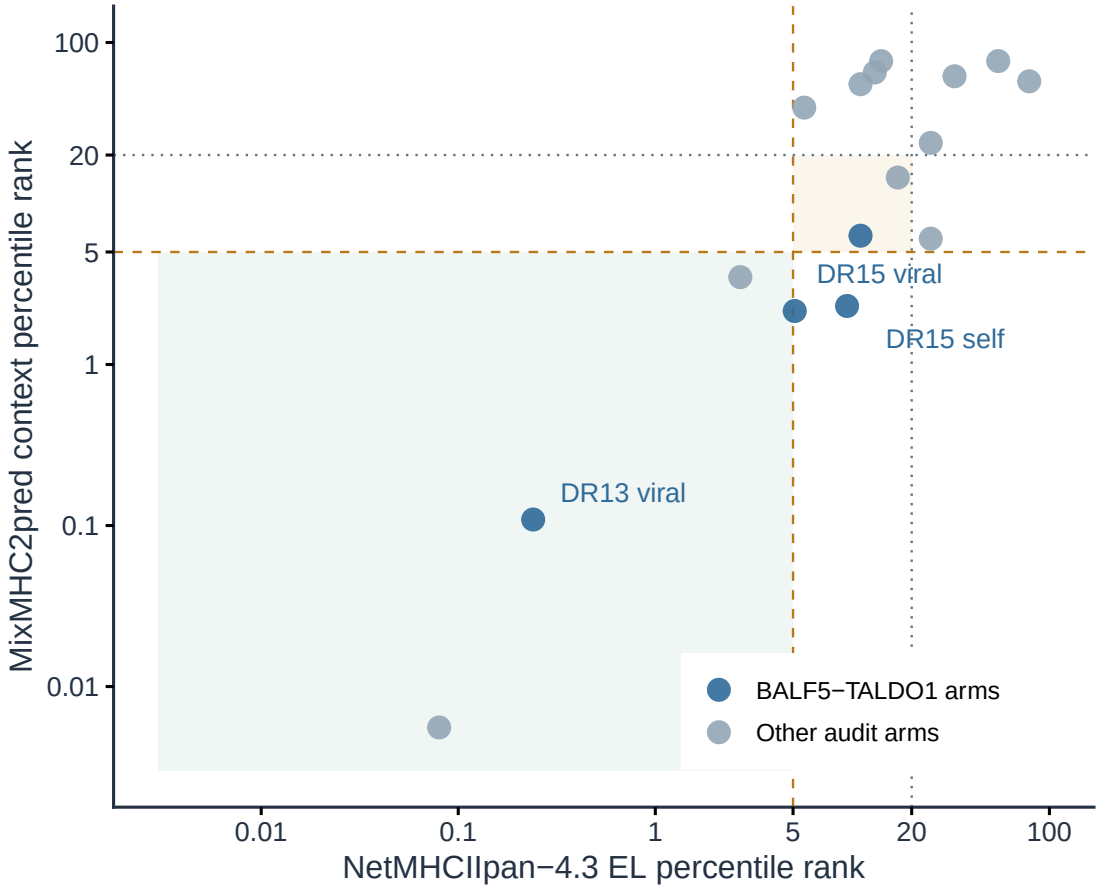


Figure S2: **Concordance of peptide-HLA class II binding-percentile predictions in the initial eight-pair audit.** Each point represents one peptide arm. NetMHCIIpan-4.3 EL and MixMHC2pred context percentile ranks are plotted on logarithmic axes; lower values indicate stronger predicted binding. Gold dashed lines mark the $\leq$ 5% full-consensus threshold, gray dotted lines mark the $\leq$ 20% lesser-support threshold, and the shaded region identifies arms below 5% in both predictors. Blue points are BALF5-TALDO1 arms; gray points are other audited arms.

3 Supplementary Tables

Table S1: Top sequence-ranked pairs within each allele.

Allele	Rank 1	Rank 2	Rank 3
*03:01	EBNA2$_{136\text{-}150}$-MOG$_{97\text{-}109}$ (0.48)	BNRF1$_{37\text{-}51}$-CLDN11$_{193\text{-}207}$ (0.44)	BALF4$_{103\text{-}116}$-CRYAB$_{1\text{-}15}$ (0.41)
08:01	EBNA1$_{87\text{-}101}$-ANO2$_{79\text{-}93}$ (0.52)	BLLF1$_{65\text{-}75}$-PLP1$_{140\text{-}152}$ (0.42)	BZLF1$_{198\text{-}210}$-MAG$_{612\text{-}626}$ (0.391)
13:03	EBNA3B$_{139\text{-}153}$-MOG$_{221\text{-}235}$ (0.49)	BNRF1$_{37\text{-}51}$-CLDN11$_{193\text{-}207}$ (0.44)	BARF1$_{1\text{-}15}$-MAG$_{1\text{-}15}$ (0.44)
15:01	EBNA1$_{87\text{-}101}$-ANO2$_{79\text{-}93}$ (0.52)	EBNA3B$_{139\text{-}153}$-MOG$_{221\text{-}235}$ (0.49)	BALF4$_{575\text{-}589}$-CNP$_{1\text{-}15}$ (0.43)

Table S2: Initial eight-candidate evidence audit.

Candidate pair HLA context	Binding V / S	Register V / S	IEDB arms	Ligand arms	Audit status
EBNA2-MOG					
HLA-DRB1*03:01	No / Yes	Yes / Yes	0	0	Hold
BARF1-CRYAB					
HLA-DRB1*03:01	No / No	No / Yes	0	0	Hold
EBNA1-ANO2					
HLA-DRB1*08:01	No / No	Yes / Yes	0	0	Hold
EBNA-LP-MOG					
HLA-DRB1*08:01	No / No	No / Yes	0	0	Hold
EBNA3B-MOG					
HLA-DRB1*13:03	No / No	No / No	0	0	Hold
BALF5-TALDO1					
HLA-DRB1*13:03	Yes / No	Yes / Yes	0	0	**Medium** priority
BALF4-CRYAB					
HLA-DRB1*15:01	Yes / No	Yes / Yes	1	1	Hold
BALF5-TALDO1					
HLA-DRB1*15:01	No / No	Yes / Yes	1	0	**Medium** priority

Note. V/S, viral/self peptide arm. Binding: both NetMHCIIpan-4.3 EL and MixMHC2pred ranks $\leq 5\%$ for that arm. Register: both predictors selected the same declared P1-P9 core. IEDB and ligand entries are numbers of peptide arms with exact-HLA supporting records (0-2); they are not numbers of studies or patients. These fields do not establish natural presentation or T-cell cross-reactivity.

Table S3: Secondary provisional computational strata in the 49-pair cohort.

Target allele	Viral-human pair	Status	Supporting factor	Limitation
DRB1*15:01	$BALF5_{627\text{-}641}$-$TALDO1_{216\text{-}230}$	Stratum A	Complete computational consensus; viral IEDB binding	Register instability; no self-arm elution record
DRB1*13:03	$BALF5_{627\text{-}641}$-$TALDO1_{108\text{-}122}$	Stratum B	Expanded binding and register criteria	No verified public record
DRB1*15:01	$EBNA1_{87\text{-}101}$-$MBP_{217\text{-}233}$	Stratum B	Declared register retained across calibrated PANDORA templates	Binding-predictor discrepancy; lower rank
DRB1*15:01	$BALF4_{103\text{-}116}$-$CRYAB_{1\text{-}15}$	Stratum C	Initial sequence similarity	Template discordance and register sensitivity
DRB1*03:01	$EBNA2_{136\text{-}150}$-$MOG_{97\text{-}109}$	Stratum C	Initial sequence rank	No same-allele PANDORA calibration
DRB1*08:01	$BHRF1_{122\text{-}133}$-$ANO2_{104\text{-}118}$	Stratum C	Initial sequence rank	No same-allele PANDORA calibration
DRB1*08:01	$EBNA1_{87\text{-}101}$-$ANO2_{79\text{-}93}$	Stratum C	Initial sequence rank	No same-allele PANDORA calibration
Various	Various	Untiered	Heterogeneous profiles	Incomplete consensus or evidence

Table S4: Constrained-null simulation results for the eight audited pairs. Ten thousand replicates were completed independently for each allele at random seed 20260914. The 95% null interval reports the 2.5th and 97.5th percentiles.

Target	Allele	Pair	Observed score (rank)	Null median	95% null interval	Monte Carlo p	BH q	maxT p
HY03_SEQ_01	DRB1*03:01	EBNA2-MOG	0.48000 (1)	-0.18519	[-0.35714, 0.28000]	0.00790	0.06319	0.83572
HY03_SEQ_02	DRB1*03:01	BARF1-CRYAB	0.27273 (13)	-0.17949	[-0.32500, 0.34375]	0.08189	0.08189	1.00000
HY08_SEQ_01	DRB1*08:01	EBNA1-ANO2	0.52174 (1)	0.00000	[-0.32000, 0.52174]	0.03420	0.07199	0.66483
HY08_SEQ_02	DRB1*08:01	EBNA-LP-MOG	0.32000 (6)	-0.11538	[-0.46154, 0.32000]	0.06539	0.07634	1.00000
HY13_SEQ_01	DRB1*13:03	EBNA3B-MOG	0.48485 (1)	-0.13514	[-0.32432, 0.48485]	0.03600	0.07199	0.82102
HY13_SEQ_02	DRB1*13:03	BALF5-TALDO1	0.28125 (13)	-0.15625	[-0.33333, 0.25806]	0.01720	0.06879	1.00000
HY15_SEQ_01	DRB1*15:01	BALF4-CRYAB	0.41176 (5)	-0.10000	[-0.30233, 0.47059]	0.05559	0.07634	0.98260
HY15_SEQ_02	DRB1*15:01	BALF5-TALDO1	0.31429 (14)	0.05714	[-0.18919, 0.41176]	0.06679	0.07634	1.00000

Sequence-score calibration only; not evidence of endogenous presentation, pMHC stability, TCR recognition, cross-reactivity, molecular mimicry, or an MS mechanism.

Table S5: Declared P1–P9 cores for the two BALF5-TALDO1 sequence-prioritized comparisons. Bold residues are the nominal HLA-facing positions P1, P4, P6, and P9; unbolded residues are the designated TCR-facing positions P2, P3, P5, P7, and P8. These are frozen V3 input cores, not experimentally mapped registers.

Allele	Peptide arm	Declared core (P1–P9)
DRB1*13:03	$BALF5_{627\text{-}641}$	**Y**HF**V**K**K**HV**H**
DRB1*13:03	$TALDO1_{108\text{-}122}$	**L**SF**D**K**D**AM**V**
DRB1*15:01	$BALF5_{627\text{-}641}$	**V**YH**F**V**K**KH**V**
DRB1*15:01	$TALDO1_{216\text{-}230}$	**I**YN**Y**Y**K**KF**S**

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